MALAPPURAM SAHODAYA APTITUDE TEST 2024-25

SCHOOL LEVEL

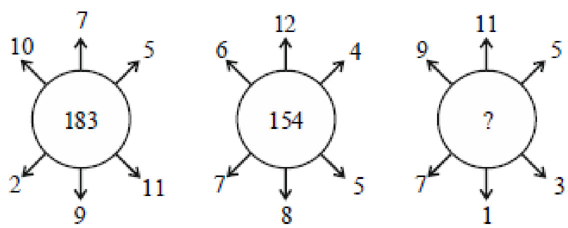
Class: 11 - PCB **Date:** **Marks: 100**

Name of the Student: ____

GENERAL INSTRUCTIONS

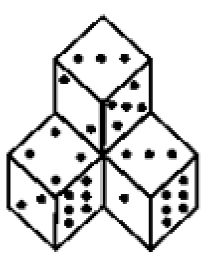
1. ***Read the questions carefully.***
2. ***Attempt all the questions.***
3. ***No negative marking.***
4. ***Answers to be marked in the OMR sheet. Darken the bubble completely. Partially darkened bubbles will not be considered.***
5. ***Question numbers 1 to 100 carries 1 mark each.***

1. Find the missing character

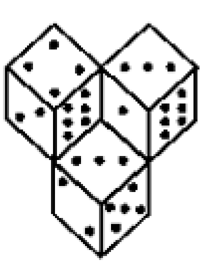


a. 86 b. 73 c. 33 d. 104

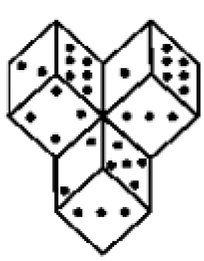
2 Find the water image of X



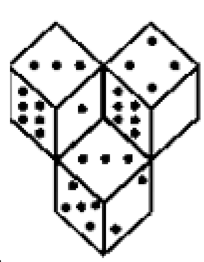
(X)



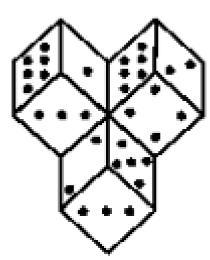
a.



b.

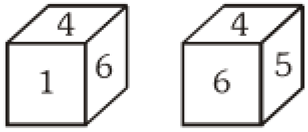


c.



d.

3 Two positions of a dice are shown below. When number '1' is on the top, what number will be at the bottom?



a. 3 b. 5 c. 2 d. 6

4.CDEF : MNOP: : UVWX:?

a. A. DEFG b. EFGH. c. GHIJ d. ABCD

5. Complete the analogy:

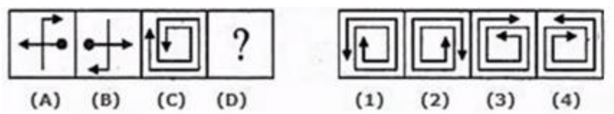
Cardiologist : Heart :: Chiropractor :

a. Ear b. Joint c. Brain d. Skull

6. Guess the next number: 19 : 361 :: 66 : ____

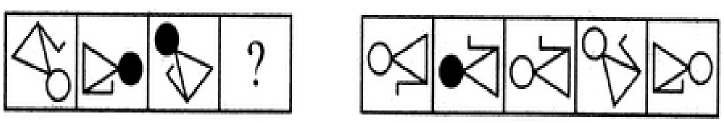
A.693 B.256 C.4356. D.5346

7. Which suitable figure will replace the question mark based on the figures given in question?



A. 1 B. 2. C. 3 D. 4

8 .Based on the figure in the question find a suitable figure from the five figures in the option



A. 1 B. 2. C. 3 D. 4

9. In a row all the persons are facing north, Rahul is 33rd from the left end and in the right side of Rahul, there are only 16 persons. Find out total number of persons in this queue?

A. 49 B. 50 C. 51 D. None of these

10. Five friends are standing in a line. Nishu is taller than Riya but shorter than Pooja. Amrita is the shortest. Riya is shorter than Nishu but taller than Nikita. Who is the second tallest?

A. Amrita. B. Pooja. C. Riya D.Nishu

11. There are 25 boys in a horizontal row. Rahul was shifted by three places towards his right side and he occupies the middle position in the row. What was his Original position from the left end of the row

A. 15. B. 14. C. 12. D.10

12. Find the odd one out: 396, 462, 572, 427, 671, 264

A. 396. B.427 C. 671 D.264

13. On 8th Dec, 2007 Saturday falls. What day of the week was it on 8th Dec, 2006 ?

A. Sunday B. Thursday. C. Tuesday D. Friday

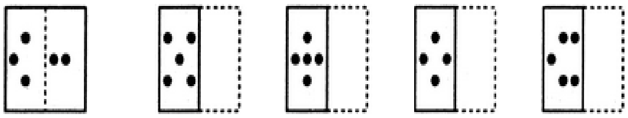
14. Clock is started at noon. By 10 minutes past 5, the hour hand has turned through:

A.145°. B.150°. C.155° D.160°

15. The average of 20 numbers is zero. Of them, at the most, how many may be greater than zero?

A.0. B.1. C.10. D.19

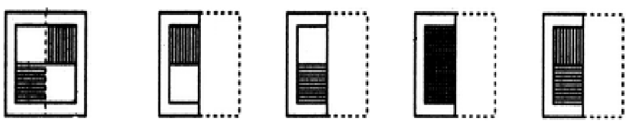
16. Find out from amongst the four alternatives as to how the pattern would appear when the transparent sheet is folded at the dotted line.



(X). (1). (2). (3). (4)

A. 1. B. 2. C. 3 D .4

17. Find out from amongst the four alternatives as to how the pattern would appear when the transparent sheet is folded at the dotted line.



(X). (1). (2). (3). (4)

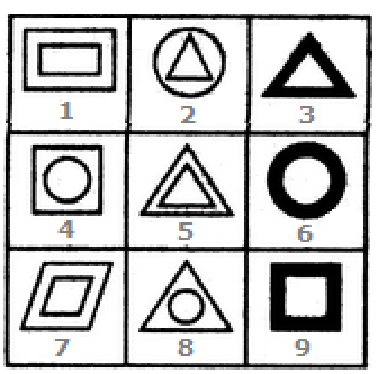
A .1. B.2. C. 3. D.4

18.Find the next term in the alphanumeric series: Z1A, X2D, V6G, T21J, R88M, P445P, ?

A.N2676P. B.N2676S. C. T2670N. D. T2676S

Group the given figures in Qn. 19 & 20 into three classes using each figure only once.

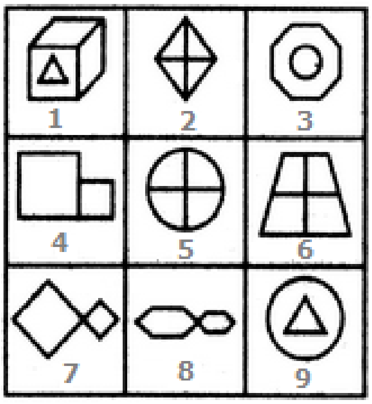
19.



A)1,5,7 ; 2,4,6 ; 3,9,8 B).1,5,7 ; 2,4,8 ; 3,6,9.

C)1,4,7 ; 2,5,8 ; 3,6,9 D)1,7,9 ; 3,5,8 ; 2,4,6

20.



a. 1,3,9 ; 2,5,6 ; 4,7,8.   b. 1,3,9 ; 2,7,8 ; 4,5,6

c. 1,2,4 ; 3,5,7 ; 6,8,9   d. 1,3,6; 2,4,8 ; 5,7,9

21. Peter is in the East of Tom, and Tom is in the North of John. Mike is in the South of John, then in which direction of Peter is Mike?

A)South-East   B)South-West   C)South.   D)North-East

22.There are four pipes used to fill a tank at a constant rate. Altogether, the four pipes are able to fill up the tank in 4 hours. The owner wishes to reduce this time taken to 2 hours. How many more pipes filling up the tank at a constant rate will they need to install?

a. 2.   b.8.   c. 4.   d.6

23.In a certain store, the profit is 320% of the cost. If the cost increases by 25% but the selling price remains constant, approximately what percentage of the selling price is the profit?

a. 30%   b. 70%.   c.100%.   d.250%

24.A vendor bought toffees at 6 for a rupee. How many for a rupee must he sell to gain 20%?

a. 3.   b. 4.   c. 5.   d.6

25.Four defensive football players are chasing the opposing wide receiver, who has the ball. Calvin is directly behind the ball carrier. Jenkins and Burton are side by side behind Calvin. Zeller is behind Jenkins and Burton. Calvin tries for the tackle but misses and falls. Burton trips. Which defensive player tackles the receiver?

a. Burton.   b. Zeller   c. Jenkins   d. Calvin

26. Select the correct meaning of the word “obfuscate”:

a) Simplify   b) Confuse   c) Clarify   d) Support

27. Which of the following is a synonym of “pernicious”?

a) Harmless   b) Beneficial   c) Destructive   d) Friendly

28. Choose the correct antonym of “capricious”:

a) Unpredictable   b) Constant   c) Erratic   d) Arbitrary

29. Which word best fits the sentence?

_The _______ politician swayed the audience with his manipulative speech._

a) candid   b) disingenuous   c) amiable   d) effusive

30. What does the term “belligerent” mean?

a) Friendly b) Hostile c) Indifferent d) Relaxed

31. Identify the correct noun form in the sentence:

_The President’s _______ was clear and concise._

a) announce b) announcer c) announcement d) announcing

32. Which sentence uses a demonstrative pronoun correctly?

a) That are my books. c) These is the way.

b) Those are your options. d) This are the result.

33. Complete the sentence using the correct verb form:

_By next year, they ________ the project._

a) will complete c) will have completed

b) will be completing d) have completed

34. Choose the correct adverb for the sentence:

_She handled the situation ______._

a) with care b) carefully c) carelessly d) careful

35. Select the correct comparative adjective:

_This test is _______ than the previous one._

a) more easier b) easier c) easy d) easiest

36. Choose the correct preposition:

_The decision depends ____ the outcome of the meeting._

a) from b) on c) in d) for

37. Complete the sentence with the correct conjunction:

She studied hard ______ she could pass the exam.

a) so that b) and c) but d) while

38. Select the correct modal verb:

He _______ take an umbrella since it’s going to rain.

a) should b) must c) can d) might

39. Which sentence correctly uses a gerund?

a) He is fond of to play football. c) He is interested to play football.

b) He enjoys playing football. d) He enjoys to play football.

40. Choose the correct participle:

The ________ tree provided much-needed shade.

a) fallen b) falling c) fall d) fell

41. Which of the following sentences is in the passive voice?

a) The teacher explains the lesson. b) The lesson is explained by the teacher.

c) The teacher will explain the lesson. d) The teacher is explaining the lesson.

42. Choose the correct indirect speech:

He said, “I have completed the work.”

a) He said that he had completed the work.
c) He said that he has completed the work.
b) He says he completed the work.
d) He said that he will complete the work.

43. Form a question from the given sentence:

They are going to the party.

a) Are they going to the party?
c) They are going to the party, right?
b) Will they be going to the party?
d) Is they going to the party?

44. Identify the type of sentence:

Please submit the report by tomorrow.

a) Declarative b) Imperative c) Interrogative d) Exclamatory

Clauses, Idioms, and Proverbs

45. Identify the subordinate clause in the sentence:

If you study hard, you will succeed.

a) If you study hard
c) Study hard
b) You will succeed
d) Succeed

46. What does the idiom "bite the bullet" mean?

a) To avoid responsibility
c) To procrastinate
b) To face a difficult situation with courage
d) To react impulsively

47. Choose the correct meaning of the proverb "The early bird catches the worm":

a) Early risers get the best opportunities.
c) Preparation is the key to success.
b) Hard work pays off in the end.
d) Opportunities are rare

48. Which of the following sentences is punctuated correctly?

a) However, I still don’t agree with you.
c) However I still, don’t agree with you.
b) However I still don’t agree with you.
d) However; I still don’t agree with you.

49. What is the purpose of a colon in the following sentence?

She brought three items: a pen, a notebook, and a calculator.

a) To introduce a list
c) To show possession
b) To separate two independent clauses
d) To express emphasis

50. Choose the correct transformation of this sentence:

She is too tired to work.

a) She is so tired that she cannot work.
d) She is tired yet working
b) She is very tired, so she can work.
c) She is tired but can work.

51. Which of the following are true about the taxonomical aid 'key'?

1. Keys are based on the similarities and dissimilarities.
2. Key is analytical in nature.
3. Keys are based on the contrasting characters in pair called Couplet.
4. Same key can be used for all taxonomic categories.
5. Each statement in the key is called Lead.

Choose the most appropriate answer from the options given below:

a. (i), (iii), (iv) and (v) only
b. (i), (ii) and (iii) only
c. (ii), (iii) and (iv) only
d. (i), (ii), (iii) and (v) only

52. Choose the wrong statement.

a. Morels and truffles are poisonous mushrooms.
b. Yeast is unicellular and useful in fermentation.
c. Penicillium is multicellular and produces antibiotics.
d. Neurospora is used in the study of biochemical genetics.

53. Which one of the following also acts as a catalyst in a bacterial cell?

a. 5S rRNA
b. snRNA
c. hnRNA
d. 23S rRNA

54. Read the following statements and choose the set of correct statements: In the members of Phaeophyceae,

A. Asexual reproduction occurs usually by biflagellate zoospores.

B. Sexual reproduction is by oogamous method only.

C. Stored food is in the form of carbohydrates which is either mannitol or laminarin.

D. The major pigments found are chlorophyll a, c and carotenoids and xanthophyll.

E. Vegetative cells have a cellulosic wall, usually covered on the outside by gelatinous coating of algin.

Choose the correct answer from the options given below:

a. A, B, C and D only
b. B, C, D and E only
c. A, C, D and E only
d. A, B, C and E only

55. Match List-I with List-II.

List - I		List - II	
A	Chondrichthyes	I	Clarias
B	Cyclostomata	II	Carcharodon
C	Osteichthyes	III	Myxine
D	Amphibia	IV	Ichthyophis

Choose the correct answer from the options given below:

a. A-II, B-IV, C-I, D-III
b. A-I, B-III, C-II, D-IV
c. A-II, B-III, C-I, D - IV
d. A-I, B-II, C-III, D-IV

56. Match List-I with List-II:

List - I		List - II	
A.	Vexillary aestivation	I.	Brinjal
B.	Epipetalous stamens	II.	Peach
C.	Epiphyllous stamens	III.	Pea
D.	Perigynous flower	IV.	Lily

Choose the correct answer from the options given below:

Options:

a. A-III, B-I, C-IV, D-II
b. A-III, B-IV, C-I, D-II
c. A-III, B-I, C-I, D-IV
d. A-II, B-I, C-IV, D-III

57. Given below are two statements:

Statement I: In a dicotyledonous leaf, the adaxial epidermis generally bears more stomata than the abaxial epidermis.

Statement II: In a dicotyledonous leaf, the adaxially placed palisade parenchyma is made up of elongated cells, which are arranged vertically and parallel to each other.

In the light of the above statements, choose the correct answer from the options given below:

a. Statement I is true but Statement II is false
b. Statement I is false but Statement II is true
c. Both Statement I and Statement II are true
d. Both Statement I and Statement II are false

58. Given below are two statements:

Statement I: Ligaments are dense irregular tissue.

Statement II: Cartilage is dense regular tissue.

In the light of the above statements, choose the correct answer from the options given below:

a. Both Statement I and Statement II are false
b. Statement I is true but Statement II is false.
c. Statement I is false but Statement II is true
d. Both Statement I and Statement II are true

59. Given below are two statements:

Statement I: Bt toxins are insect group specific and coded by a gene cry IAc.

Statement II: Bt toxin exists as inactive protoxin in B. thuringiensis. However, after ingestion by the insect the inactive protoxin gets converted into active form due to acidic pH of the insect gut.

In the light of the above statements, choose the correct answer from the options given below:

a. Both Statement I and Statement II are true
b. Both Statement I and Statement II are false
c. Statement I is true but Statement II is false
d. Statement I is false but Statement II is true

60. Match the columns and identify the correct option.

Column I	Column 1I
A. Thylakoids	i. Disc-shaped sacs in Golgi apparatus
B. Cristae	ii. Condensed structure of DNA
C. Cisternae	iii. Flat membranous sacs in stroma
D. Chromatin	iv. Infoldings in mitochondria

a. A-(iii), B-(i). C-(iv), D-(ii)
b. A-(iii), B-(iv), C-(ii), D-(i)
c. A-(iv), B-(iii), C-(i). D-(ii)
d. A-(iii), B-(iv), C-(i), D-(ii)

61. A phosphoglyceride is always made up of

a. a saturated or unsaturated fatty acid esterified to a glycerol molecule to which a phosphate group is also attached
b. a saturated or unsaturated fatty acid esterified to a phosphate group which is also attached to a glycerol molecule
c. only a saturated fatty acid esterified to a glycerol molecule to which a phosphate group is also attached
d. only an unsaturated fatty acid esterified to a glycerol molecule to which a phosphate group is also attached.

62. Match the stages of meiosis in column I to their characteristic features in column II and select the correct option using the codes given below.

Column I	Column II
A. Pachytene	(i) Pairing of homologous chromosomes
B. Metaphase I	(ii) Terminalisation of chiasmata
C. Diakinesis	(iii) Crossing-over takes place
D. Zygotene	(iv) Chromosomes align at equatorial plate

a. A-(iii), B-(iv), C-(ii), D-(i)
b. A-(i), B-(iv), C-(ii), D-(iii)
c. A-(ii), B-(iv), C-(iii), D-(i)
d. A-(iv), B-(iii), C-(ii), D-(i)

63. Two cells A and Bare contiguous. Cell A has osmotic pressure 10 atm, turgor pressure 7 atm and diffusion pressure deficit 3 atm. Cell B has osmotic pressure 8 atm, turgor pressure 3 atm and diffusion pressure deficit 5 atm. The result will be

a. no movement of water
b. equilibrium between the two
c. movement of water from cell A to B
d. movement of water from cell B to A

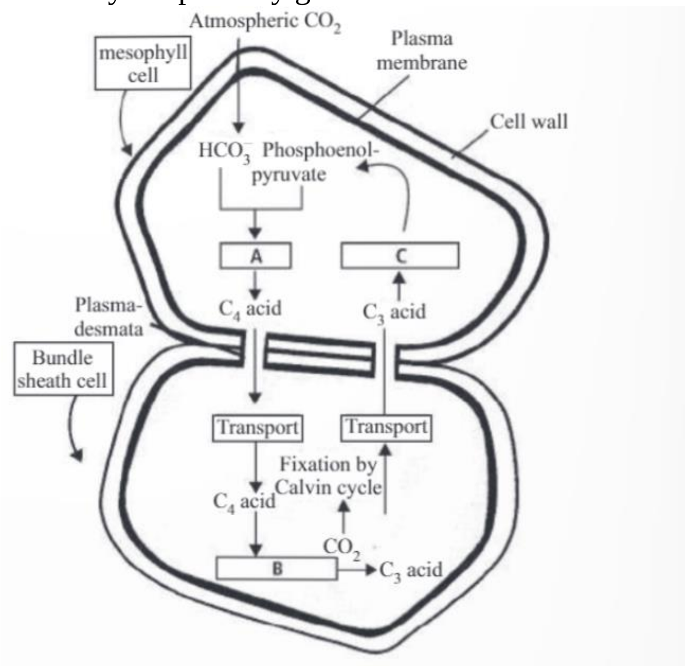
64. Match the following concerning essential elements and their functions in plants.

Column-I	Column-II
(A) Iron	(i) Photolysis of water
(B) Zinc	(ii) Pollen germination
(C) Boron	(iii) Required for chlorophyll biosynthesis
(D) Manganese	(iv) IAA biosynthesis

Select the correct option.

a. (A)-(ii), (B)-(i), (C)-(iv), (D)-(iii)
b. (A)-(iv), (B)-(iii), (C)-(ii), (D)-(i)
c. (A)-(iii), (B)-(iv), (C)-(ii), (D)-(i)
d. (A)-(iv), (B)-(i), (C)-(ii), (D)-(iii)

65. Study the pathway given below.



In which of the following options correct words for all the three blanks A, B and C are indicated?

a. A-Decarboxylation, B-Reduction, C-Regeneration
b. A-Fixation, B-Transamination, C-Regeneration
c. A-Fixation, B-Decarboxylation, C-Regeneration
d. A-Carboxylation, B-Decarboxylation, C-Reduction

66. Which statement is wrong for Krebs' cycle?

a. There is one point in the cycle where FAD^+ is reduced to $FADH_2$.
b. During conversion of succinyl CoA to succinic acid, a molecule of GTP is synthesised.
c. The cycle starts with condensation of acetyl group (acetyl CoA) with pyruvic acid to yield citric acid.
d. There are three points in the cycle where NAD^+ is reduced to $NADH^+ + H^+$.

67. Dr. F. Went noted that if coleoptile tips were removed and placed on agar for one hour, the agar would produce a bending when placed on one side of freshly-cut coleoptile stumps. Of what significance is this experiment?

a. It made possible the isolation and exact identification of auxin.
b. It is the basis for quantitative determination of small amounts of growth-promoting substances.
c. It supports the hypothesis that IAA is auxin.
d. It demonstrated polar movement of auxins.

68. Match the following structures with their respective location in organs.

(A) Crypts of Lieberkuhn	(i) Pancreas
(B) Glisson's Capsule	(ii) Duodenum
(C) Islets of Langerhans	(iii) Small intestine
(D) Brunner's Glands	(iv) Liver

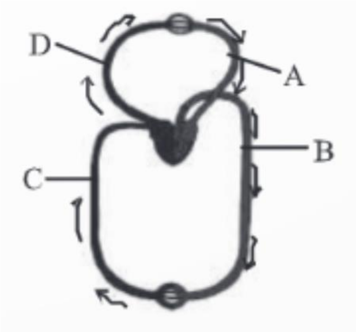
Select the correct option from the following:

a. (A)-(iii), (B)-(ii), (C)-(i), (D)-(iv)
b. (A)-(iii), (B)-(i), (C)-(ii), (D)-(iv)
c. (A)-(ii), (B)-(iv), (C)-(i), (D)-(iii)
d. (A)-(iii), (B)-(iv), (C)-(i), (D)-(ii)

69. Lungs do not collapse between breaths and some air always remains in the lungs which can never be expelled because

a. there is a negative pressure in the lungs
b. there is a negative intrapleural pressure pulling at the lung walls
c. there is a positive intrapleural pressure
d. pressure in the lungs is higher than the atmospheric pressure.

70. The given figure shows schematic plan of blood circulation in humans with labels A to D. Identify the label and give its functions.



a. C - Vena cava - Takes blood from body parts to right auricle, PCO_2 = 45 mm Hg
b. D - Dorsal aorta Takes blood from heart to - body parts, pO2 = 95 mm Hg
c. A - Pulmonary vein - Takes impure blood from body parts, pO2 = 60 mm Hg
d. B - Pulmonary artery - Takes blood from heart to lungs, pO2 = 90 mm Hg

71. A force F is applied onto a square plate of side L . If the percentage error in determining L is 2% and that in F is 4%, the permissible percentage error in determining the pressure is

a. 2%
b. 4%
c. 6%
d. 8%

72. The radius of gyration about an axis through the center of a hollow sphere with external radius a and internal radius b is

a. $\sqrt{\frac{2}{5}\frac{(a^3-b^3)}{(a^5-b^5)}}$
b. $\sqrt{\frac{1}{4}\frac{(a^4-b^4)}{(a^2-b^2)}}$
c. $\sqrt{\frac{1}{2}\frac{(a^5-b^5)}{(a^3-b^3)}}$
d. $\sqrt{\frac{2}{5}\frac{(a^5-b^5)}{(a^3-b^3)}}$

73. A ball dropped from a point A falls down vertically to C, through the midpoint B. The descending time from A to B and that from A to C are in the ratio

a. $1:1$
b. $1:\sqrt{2}$
c. $1:3$
d. $1:2$

74. A cricket ball is hit at an angle of $30°$ to the horizontal with a kinetic energy E. Its kinetic energy when it reaches the highest point is

a. $\frac{E}{2}$
b. 0
c. $\frac{3E}{4}$
d. $\frac{2E}{3}$

75. A body hanging from a massless spring stretches it by 3 cm on earth's surface. At a place 800 km above the earth's surface, the same body will stretch the spring by (Radius of earth $= 6400$ km)

a. $\left(\frac{34}{27}\right)$ cm
b. $\left(\frac{64}{27}\right)$ cm
c. $\left(\frac{27}{64}\right)$ cm
d. $\left(\frac{27}{34}\right)$ cm

76. In a horizontal pipe of non-uniform cross-section, water flows with a velocity of 1 ms^{-1} at a point where the diameter of the pipe is 20 cm . The velocity of water (in ms^{-1}) at a point where the diameter of the pipe is 5 cm is

a. 16
b. 24
c. 8
d. 32

77. Two equal masses hung from two massless springs of spring constants k_1 and k_2 have equal maximum velocity when executing simple harmonic motion. The ratio of their amplitudes is

a. $\left(\frac{k_1}{k_2}\right)^{1/2}$ b. $\left(\frac{k_1}{k_2}\right)$ c. $\left(\frac{k_2}{k_1}\right)$ d. $\left(\frac{k_2}{k_1}\right)^{1/2}$

78. The harmonic mode which resonates with a closed pipe of length 22 cm , when excited by a 1875 Hz source and the number of nodes present in it respectively are (velocity of sound in air $= 330\ \mathrm{ms}^{-1}$)

a. 1^{st} , 1 c. 3^{rd} , 2
b. 5^{th} , 3 d. 5^{th} . 4

79. Displacement, x (in meters), of a body of mass 1 kg as a function of time, t, on a horizontal smooth surface is given as $x = 2t^2$. The work done in the first one second by the external force is

a. 1 J c. 8 J
b. 2 J d. 4 J

80. A rubber cord of density d, Young's modulus Y and length L is suspended vertically. If the cord extends by a length 0.5 L under its own weight, then L is

a. $\frac{Y}{2dg}$ c. $\frac{2Y}{dg}$
b. $\frac{Y}{dg}$ d. $\frac{dg}{2Y}$

81. In a thermodynamic system, Q represents the energy transferred to or from a system by heat and W represents the energy transferred to or from a system by work.

I. $Q > 0$ and $W = 0$ III. $W > 0$ and $Q = 0$
II. $Q < 0$ and $W = 0$ IV. $W < 0$ and $Q = 0$

Which of the above will lead to an increase in the internal energy of the system?

a. I only c. I and IV only
b. II only d. II and III only

82. A uniform copper rod of 50 cm length is insulated on the sides, and has its ends exposed to ice and steam respectively. If there is a layer of water 1 mm thick at each end, the temperature gradient (in $^\circ\mathrm{Cm}^{-1}$) in the bar is (Assume that the thermal conductivity of copper is $400\mathrm{Wm}^{-1}\ \mathrm{K}^{-1}$ and water is $0.4\mathrm{Wm}^{-1}\ \mathrm{K}^{-1}$)

a. 60 c. 50
b. 40 d. 55

83. A spherically uniform planet of mass 8×10^{24} kg and of radius 6×10^{6} m is orbiting around the Sun. The escape velocity for the planet is close to (Take $G = 6 \times 10^{-11}\ \mathrm{N-m^2/kg^2}$)

a. 11.2 km/s c. 4 km/s
b. 16 km/s d. 12.6 km/s

84. A boy is standing on a weighing machine inside a lift. When the lift goes upwards with acceleration $\frac{g}{4}$, the machine shows the reading 50 kg . wt. When the lift goes downward with acceleration $\frac{g}{4}$, the reading of the machine in kg . wt. would be

a. 50 c. 45.5
b. 30 d. 62.5

85. If n bullets each of mass m are fired with a velocity v per second from a machine gun, the force required to hold the gun in position is

a. mnv b. $\frac{mv}{n^2}$ c. $\frac{mv}{n}$ d. n^2mv

86. 4.28 g of NaOH is dissolved in water and the solution is made to 250 cc. What will be the molarity of the solution?

(a) $0.615\ \mathrm{mol\ L^{-1}}$ (c) $0.99\ \mathrm{mol\ L^{-1}}$
(b) $0.428\ \mathrm{mol\ L^{-1}}$ (d) $0.301\ \mathrm{mol\ L^{-1}}$

87. Select the correct statements from the following:

A. Atoms of all elements are composed of two fundamental particles.

B. The mass of electron is 9.10939×10^{-31} kg.
C. All the isotopes of a given element show same chemical properties.
D. Protons and electrons are collectively known as nucleons.
E. Dalton's atomic theory, regarded the atom as an ultimate particle of matter.
Choose the correct answer from the options given below:

(a) A and E only
(b) B, C and E only
(c) A, B and C only
(d) C, D and E only

88. The size of ionic species is correctly given in order

(a) $Cl^{7+} > Si^{4+} > Mg^{2+} > Na^{+}$
(b) $Na^{+} > Mg^{2+} > Si^{4+} > Cl^{7+}$
(c) $Na^{+} > Mg^{2+} > Cl^{7+} > Si^{4+}$
(d) $Cl^{7+} > Na^{+} > Mg^{2+} > Si^{4+}$

89. In which one of the following the central atom is sp^3 hybridised?

(a) NH_4^+
(b) BF_3
(c) SF_6
(d) PCl_5

90. Which of the following pairs have identical bond order?

(a) CN^- and NO^+
(b) CN^- and O_2^-
(c) CN^- and CN^+
(d) NO^+ and O_2^-

91. In which one of the following reactions, entropy decreases?

(a) Sodium chloride is dissolved in water
(b) Water is heated from 303 K to 353 K
(c) Sodium bicarbonate is decomposed to $Na_2CO_3(s), CO_2($ g and $H_2O(g)$
(d) Water crystallises into ice
(e) Dihydrogen molecule is decomposed into hydrogen atoms

92. Under isothermal conditions, a gas at 300 K expands from 0.1 L to 0.25 L against a constant external pressure of 2 bar. The work done by the gas is [Given that 1 Lbar $=100$ J]

(a) 30 J
(b) -30 J
(c) 5 kJ
(d) 25 J

93. In which one of the following equilibria Δn_g value is zero?

(a) $2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$
(b) $Ni(s) + 4CO(g) \rightleftharpoons Ni(CO)_4(g)$
(c) $CO_2(g) + C(s) \rightleftharpoons 2CO(g)$
(d) $H_2(g) + Br_2(g) \rightleftharpoons 2HBr(g)$

94. If the ionic product of $M(OH)_2$ is 5×10^{-10}, then the molar solubility of $M(OH)_2$ in 0.1 M NaOH is

(a) 5×10^{-12}M
(b) 5×10^{-10}M
(c) 5×10^{-8}M
(d) 5×10^{-16}M

95. Which of the following is not a characteristic of equilibrium:

(a) Rate is equal in both direction
(b) Measurable quantities are constant at equilibrium
(c) Equilibrium occurs in reversible condition
(d) Equilibrium occurs only in open vessel at constant temperature

96. The oxidation number of iron in Fe_3O_4 is

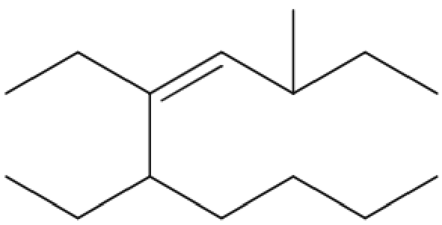
(a) +2
(b) +3
(c) +8/3
(d) +2/3

97. In the reaction,

$$3Br_2 + 6CO_3^{2-} + 3H_2O \rightarrow 5Br^- + BrO_3^- + 6HCO_3^-$$

(a) Bromine is oxidised and carbonate is reduced
(b) Bromine is reduced and water is oxidised
(c) Bromine is neither reduced nor oxidised
(d) Bromine is both reduced and oxidised

98. The correct IUPAC name of the compound is



(a) 3-heptyl-5-methylhept-3-ene
(b) 5,6-diethyl-3-methyldec-4-ene
(c) 5-butyl-3-methyloct-4-ene
(d) 8-methyl-3-propylhex-3-ene.

99. The number of sigma (σ) and pi(π) bonds in pent-2-en-4-yne is

(a) 13σ bonds and no π bond
(b) 10σ bonds and 3π bonds
(c) 8σ bonds and 5π bonds
(d) 11σ bonds and 2π bonds.

100. A tertiary butyl carbocation is more stable than a secondary butyl carbocation because of which of the following?
(a) $-I$ effect of $-CH_3$ groups
(b) $+R$ effect of $-CH_3$ groups
(c) $-R$ effect of $-CH_3$ groups
(d) Hyperconjugation

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